

Nitin Mohan

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PROFESSIONAL SUMMARY

Full-stack software engineer with production-scale experience in distributed systems, GenAI, and cloud infrastructure. Reduced GraphQL p95 latency 37% across a 50M+ record dataset and cut average response times 74% for enterprise healthcare clients. Independently architected and deployed a RAG pipeline on Google Cloud Run, contributing to Accenture's 2024 Google Cloud ML Partner of the Year award. Proficient across the full stack with hands-on experience using AI productivity tools to accelerate development workflows.

EDUCATION

Master of Science in Computer Science New York University	Expected May 2027 Brooklyn, New York
Relevant Coursework: Algorithms, Cloud Computing and Big Data, Machine Learning, Internet Security, Software Engineering, Data Engineering	
Bachelor of Technology in Computer Science Amity University	Jul. 2018 – May 2022 Uttar Pradesh, India
Relevant Coursework: Data Structures, Databases, Web Development, Operating Systems, 3 independent research studies	

WORK EXPERIENCE

Software Engineer - Full Stack LLM Development Accenture	Aug. 2022 – Jul. 2025 Bengaluru, India
- Eliminated 500+ critical SAST vulnerabilities across 8 applications by engineering middleware that resolved entire classes of XSS, SQLi, and CSRF issues, codifying fixes into Semgrep and SonarQube rules adopted as project-wide security standards. - Diagnosed a large-scale GraphQL N+1 bottleneck throttling a 50M+ record dataset at 427 RPS peak load using Datadog APM tracing, replaced per-field resolver queries with DataLoader batching, cutting per-request database calls from 150+ to under 15 and p95 latency by 37%. - Engineered a generative AI assistant for a client PoC on Google Cloud Run, integrating Vertex AI APIs with a React frontend to stream LLM responses grounded in PDF-extracted content, contributing to Accenture's 2024 Google Cloud ML Partner of the Year award. - Designed a Redis caching layer for a distributed healthcare document system serving 6k+ daily requests across NA/EU clients, achieving a 73% cache hit ratio and cutting average response time by 74% by eliminating redundant OCR reprocessing. - Led a frontend modernization initiative migrating 140+ legacy class-based React components to React 17 with Hooks and centralized Redux state, eliminating unnecessary re-renders and reducing page load time from 4.2s to 2.8s (33% faster). - Mentored 6 engineers to ticket-independent productivity within their first sprint and led React and D3.js knowledge-sharing sessions for 80-100 employees across engineering, QA, and business teams.	
Software Engineer Intern - Backend Infrastructure Ernst & Young	May 2021 – Aug. 2021 Remote
- Architected a distributed Apache Kafka streaming pipeline on AWS ingesting 1,842 daily industrial IoT sensor signals across 3 topics, enabling high-throughput delivery to multiple downstream consumer groups via MQTT broker integration. - Cut KPI dashboard query latency by 85% from 250ms to 38ms by engineering a Redis caching layer in front of Apache Spark aggregations using Python and SQL, reducing overhead on high-frequency operational metrics. - Eliminated silent data loss in the ingestion pipeline by implementing real-time alerting on consumer lag and ingestion failures, routing 70 malformed signals daily (3.8% of total throughput) to a dead-letter queue instead of dropping them.	

PROJECTS

SafeZone <i>React 19, Vite, TailwindCSS, Mapbox GL JS, AWS Cognito, Cloudinary, Make.com</i>	Link
- Engineered a geospatial rerouting algorithm computing flanking waypoints around incident clusters using bearing offsets and point-to-line distance checks (Turf.js), reconverging on a collision-free path within a bounded 3-iteration loop before falling back to the direct route. - Built a network-level mock API harness using Playwright's page.route, intercepting 5 REST resources and replicating real CRUD status semantics in memory, decoupling 4 end-to-end specs from any live backend dependency in CI. - Provisioned infrastructure as code using CloudFormation with IAM least-privilege roles and implemented dual-mode authentication supporting both production Cognito flows and local demo environments.	
RAG Paper Assistant <i>Python, FastAPI, Google Cloud Run, Pinecone, OpenRouter, Firestore, pdfminer.six</i>	Link
- Architected and deployed an end-to-end RAG pipeline covering document chunking, embedding, vector-based information retrieval, and grounded LLM response generation on Google Cloud Run and Firebase Hosting, with a fallback path to raw-text context on empty retrievals. - Enforced multi-tenant data isolation across two independent layers (Pinecone namespace-per-user and Firestore security rules), preventing cross-user data leakage without server-side authorization checks. - Designed a query routing engine integrating multiple LLMs via OpenRouter that selects between document-grounded and conversational response modes using natural language processing signals, with real-time chat history persisted in Firestore.	

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Go, Java, C#, C++, SQL, HTML, CSS, Bash, C
Frameworks & APIs: React, React Native, FastAPI, Java Spring Boot, Node.js, Angular, REST APIs, WebSocket, OAuth 2.0, JWT, RBAC
Databases & Caching: PostgreSQL, Redis, DynamoDB, TimescaleDB, CosmosDB, MySQL, OpenSearch, SQLite
Cloud & Infra: AWS (Lambda, SQS, SNS, SES, API Gateway, Cognito, S3, EC2), Azure (AKS, Event Hubs, SignalR), GCP, Docker, Kubernetes, Kafka, Terraform, Helm, Linux/Unix
DevSecOps & Platform: GitHub Actions, CI/CD, Prometheus, Semgrep, SonarQube, Unit Testing, Integration Testing, Automation Testing, Mobile Development
AI & Machine Learning: LLM integration, RAG pipelines, natural language processing, information retrieval, LangChain, LangSmith, SageMaker, agentic AI, Z-Score/EWMA anomaly detection, AI productivity tools (GitHub Copilot, Gemini)